

Nutrition Assistant

AI-Powered Personalized Nutrition and Diet Recommendation System

Project Overview

Nutrition Assistant is a MERN Stack web application that helps users manage their daily nutrition through BMI calculation, calorie estimation, water intake recommendations, meal tracking, food nutrition search, personalized diet plans, and progress analytics. The application provides a clean dashboard with interactive charts and secure user authentication.

Technology Stack

Frontend: React.js, HTML, CSS, JavaScript

Backend: Node.js, Express.js

Database: MongoDB

Authentication: JWT

Charts: Chart.js

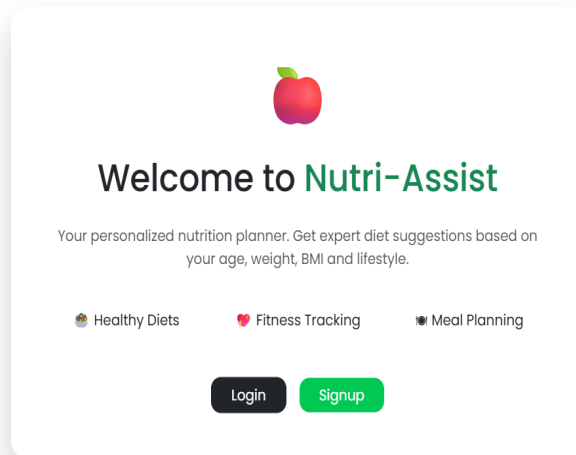
Version Control: Git & GitHub

Key Features

- User Registration and Login
- Secure Authentication
- Dashboard with Health Information
- BMI Calculator
- Daily Calorie Goal
- Water Intake Recommendation
- Protein Goal Calculation
- Meal Tracker
- Food Nutrition Search
- Personalized Diet Plan
- Profile Management
- Daily and Weekly Progress Charts

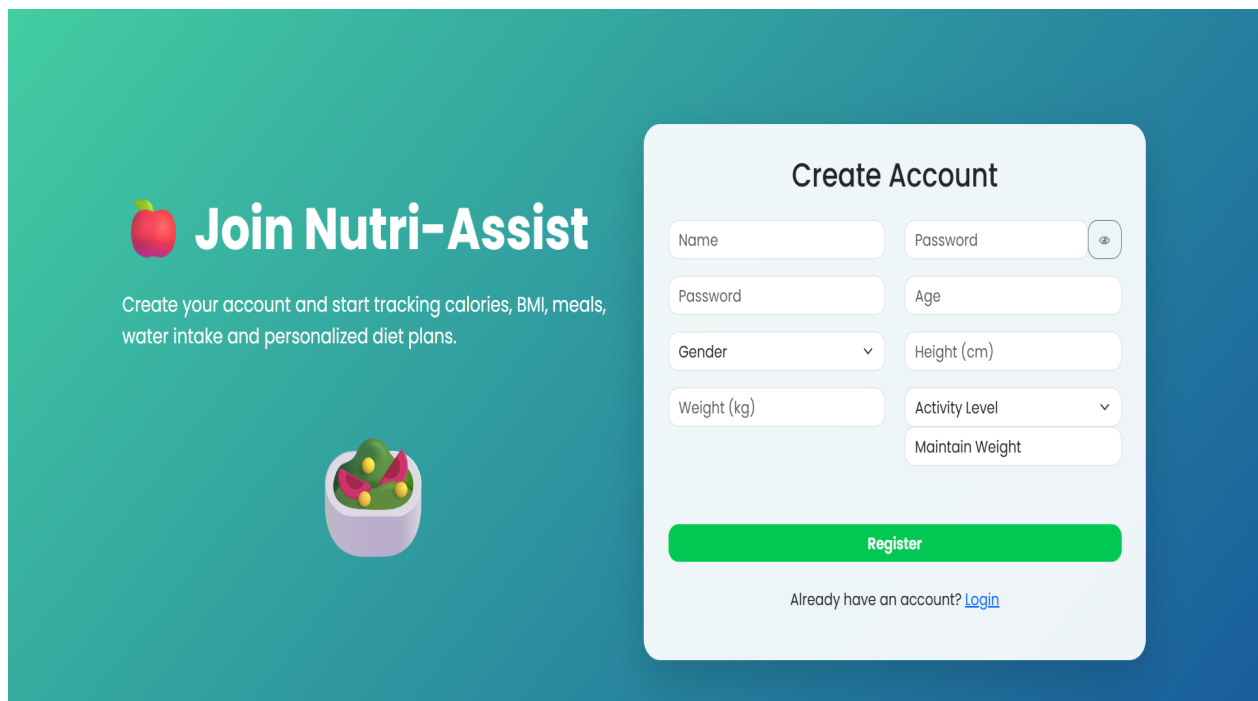
Dashboard

This screen demonstrates the **Dashboard** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



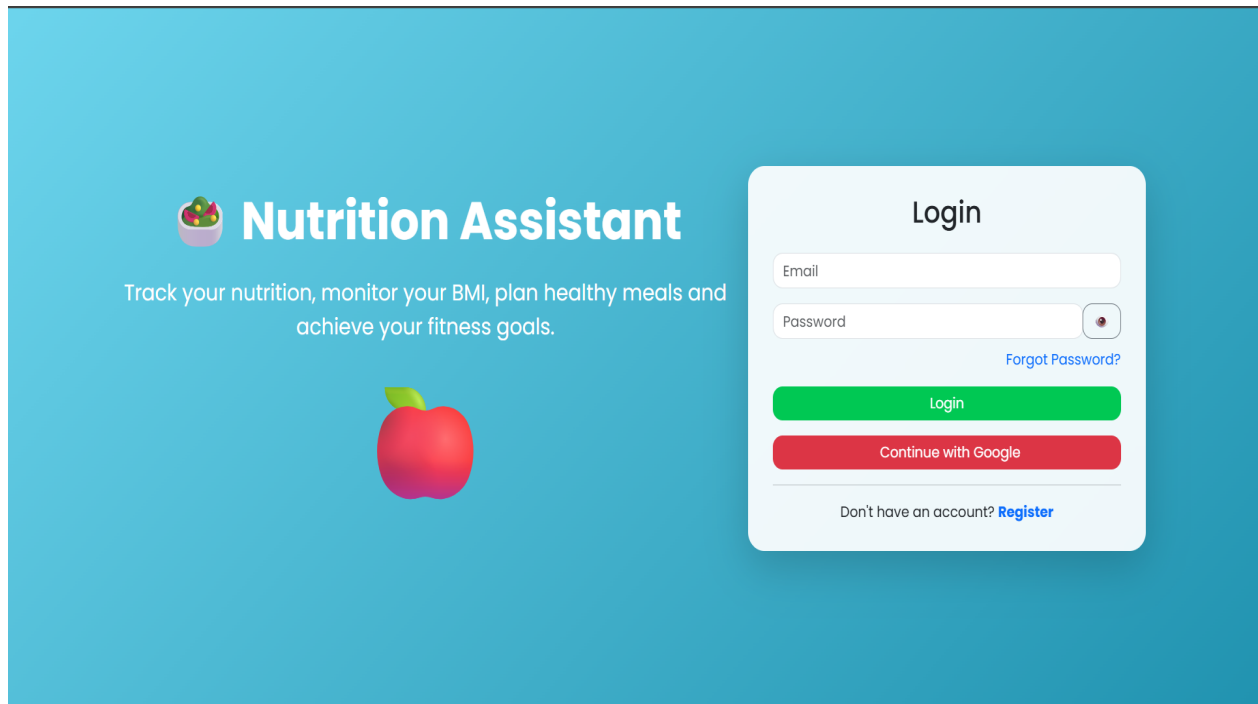
Health Information & Nutrition Summary

This screen demonstrates the **Health Information & Nutrition Summary** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



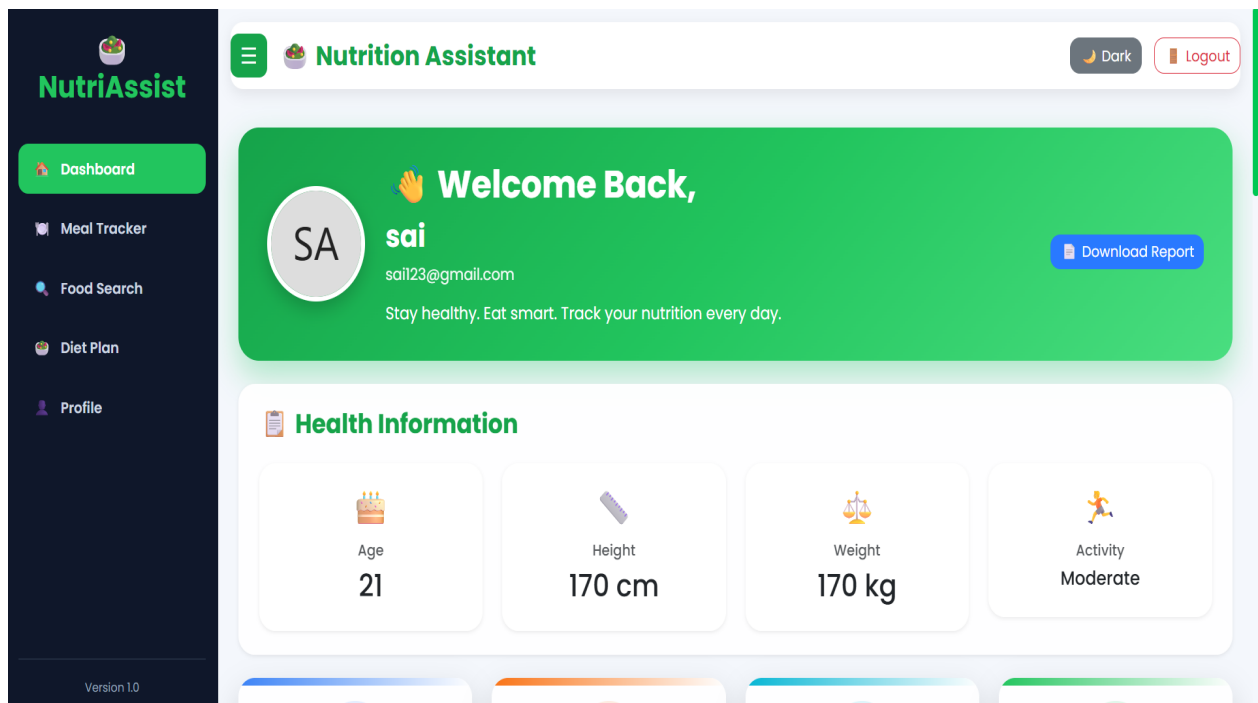
Meal Tracker

This screen demonstrates the **Meal Tracker** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



Food Nutrition Search

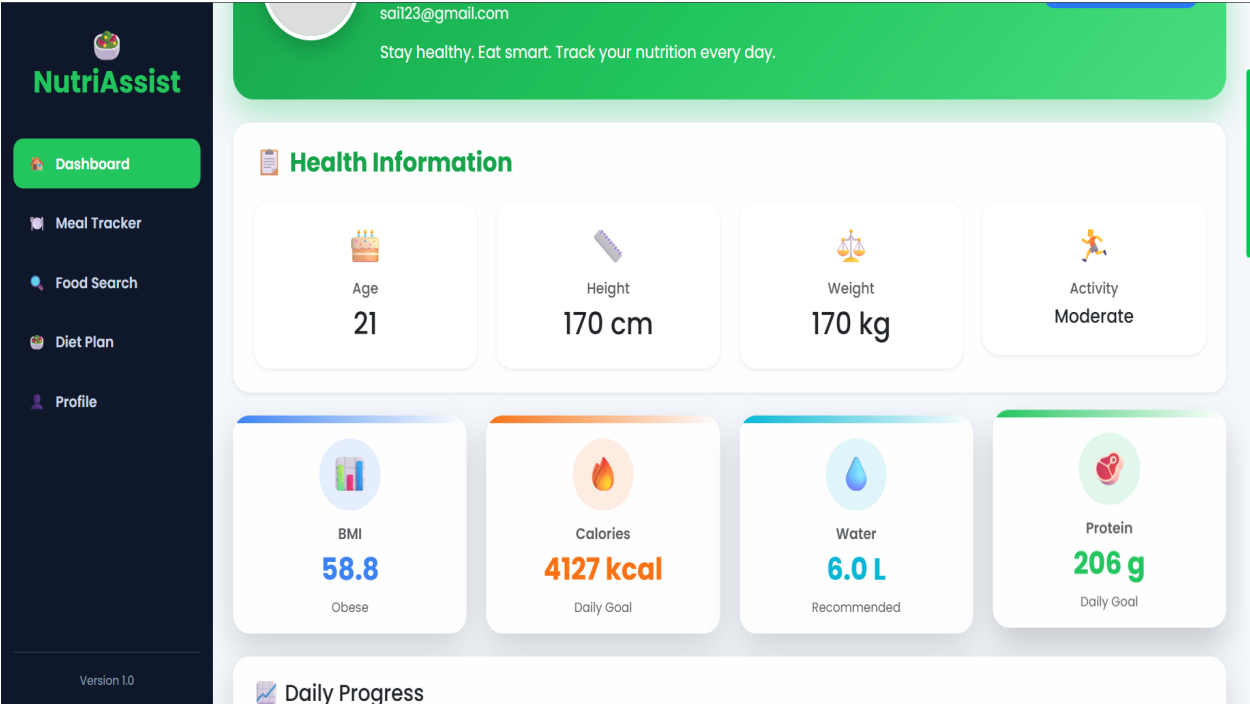
This screen demonstrates the **Food Nutrition Search** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



Personalized Diet Plan

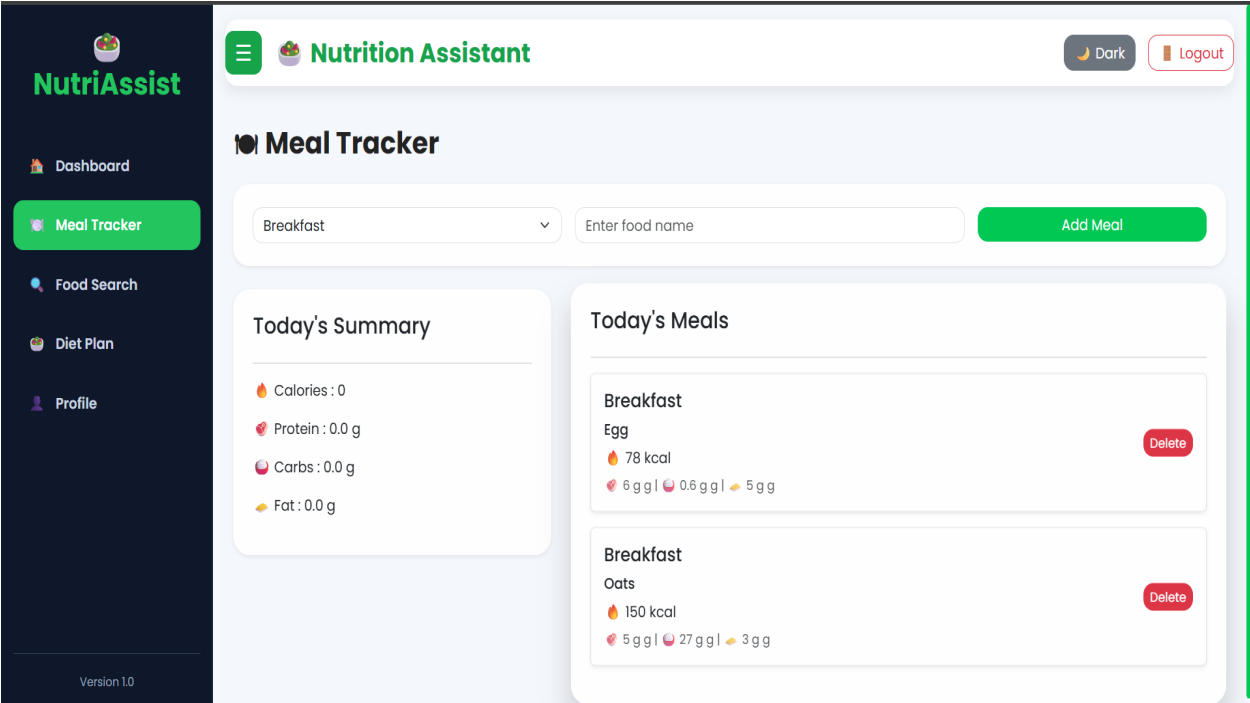
This screen demonstrates the **Personalized Diet Plan** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user

experience.



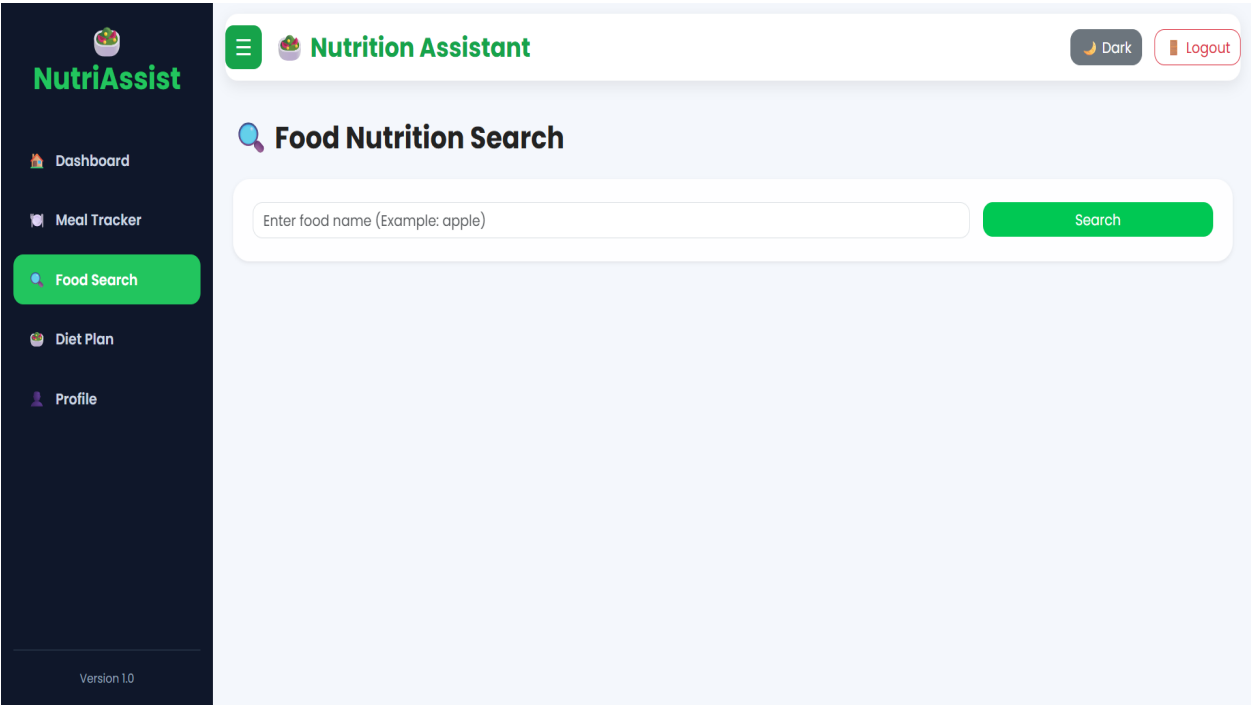
Profile Management

This screen demonstrates the **Profile Management** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



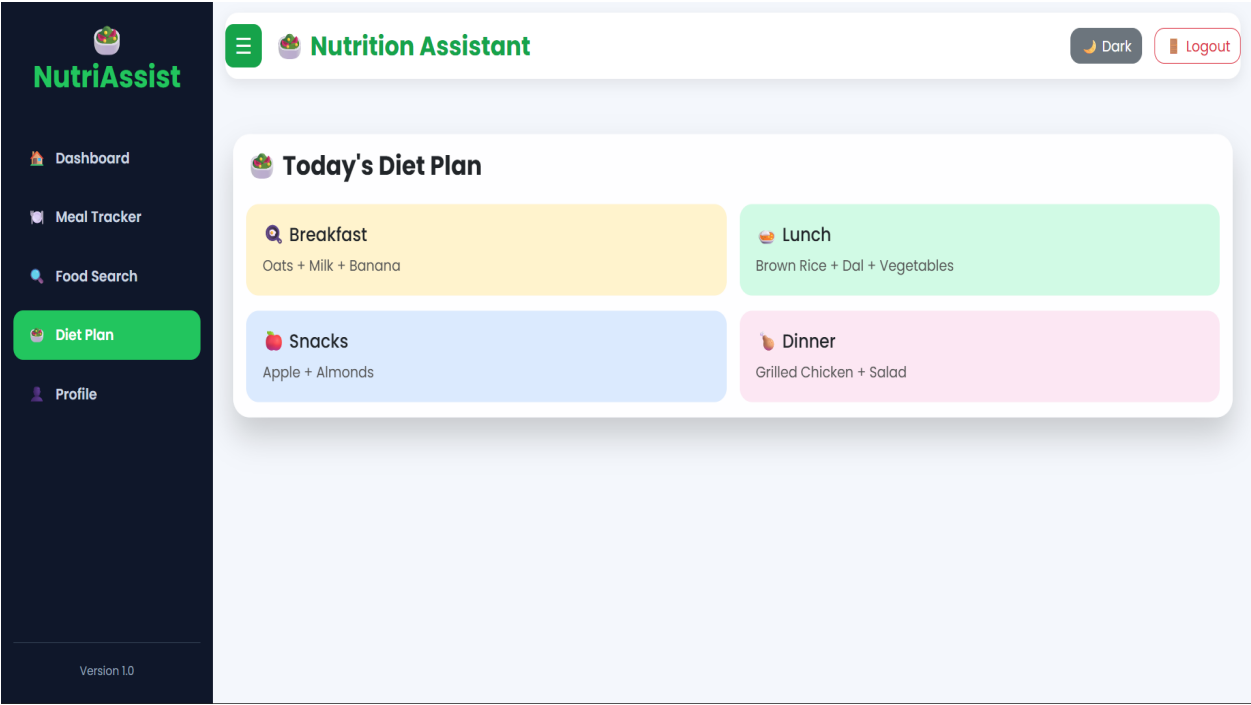
Daily Progress

This screen demonstrates the **Daily Progress** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



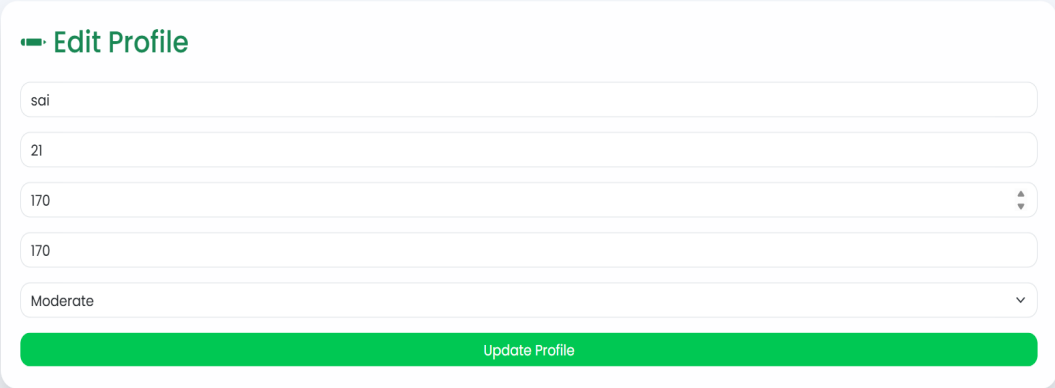
Calories Progress

This screen demonstrates the **Calories Progress** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



Weekly Calories Chart

This screen demonstrates the **Weekly Calories Chart** module of the Nutrition Assistant application. It contributes to personalized health monitoring and improves the overall user experience.



The screenshot shows a web application interface with a light blue background. A white card with rounded corners and a subtle shadow is centered. At the top left of the card is a green icon of a person and the text 'Edit Profile' in green. Below this are five input fields: a text field containing 'sai', a text field containing '21', a text field containing '170' with a small up/down arrow icon on the right, another text field containing '170', and a dropdown menu showing 'Moderate' with a downward arrow icon. At the bottom of the card is a solid green button with the text 'Update Profile' in white.

Advantages

- User-friendly interface
- Personalized nutrition recommendations
- Health monitoring in one dashboard
- Responsive design
- Secure authentication
- Easy meal tracking and nutrition analysis

Future Enhancements

- AI-based meal recommendations
- Barcode scanner
- Image-based food recognition
- Wearable integration
- Mobile application
- Advanced analytics

Conclusion

Nutrition Assistant demonstrates the practical implementation of the MERN stack by integrating authentication, database management, REST APIs, nutrition analysis, and data visualization into a single web application. The project helps users make informed dietary decisions while tracking their overall health progress.